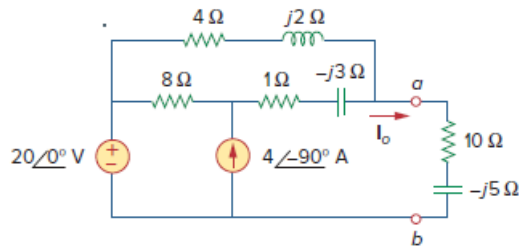


Problem

Determine the Norton equivalent of the circuit in Fig. 10.30 as seen from terminals a - b . Use the equivalent to find I_o .

Figure 10.30:



Step-by-step solution

Step 1 of 8

Refer to Figure 10.30 in the textbook.

Draw the modified circuit with the loop current notations.

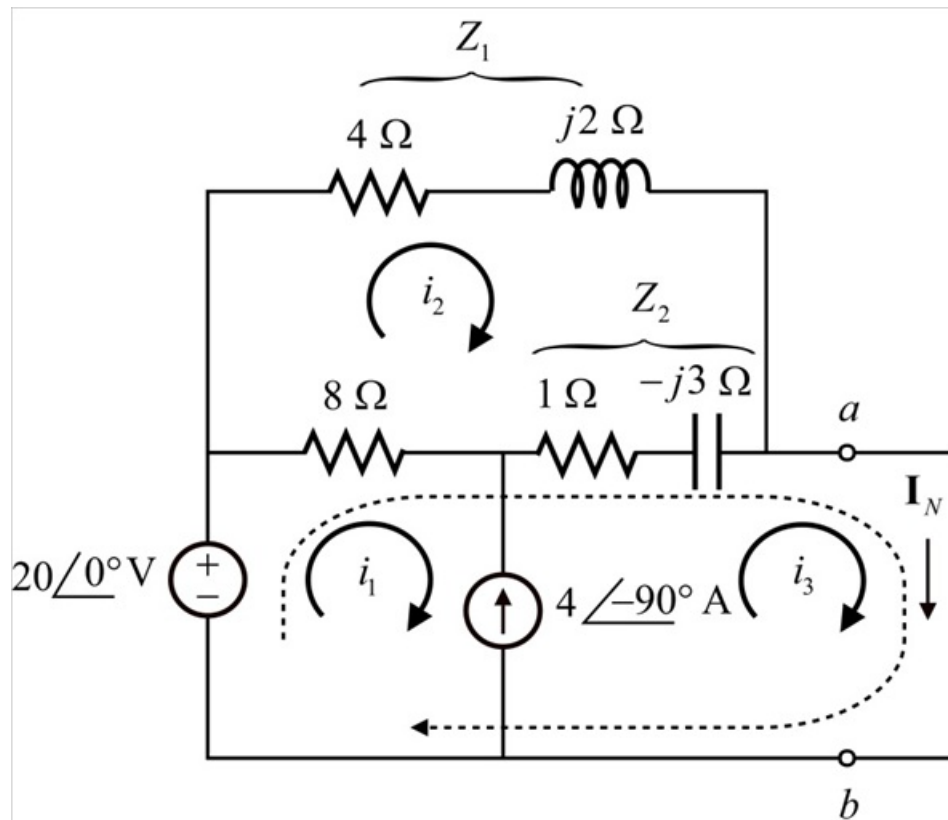


Figure 1

Step 2 of 8

Consider the following impedances:

$$Z_1 = 4 + j2$$

$$Z_2 = 1 - j3$$

Apply Kirchhoff's voltage law for the mesh 2.

$$8(i_2 - i_1) + Z_1 i_2 + Z_2 (i_2 - i_3) = 0$$

$$8(i_2 - i_1) + (4 + j2)i_2 + (1 - j3)(i_2 - i_3) = 0$$

$$-8i_1 + (8 + 4 + j2 + 1 - j3)i_2 - (1 - j3)i_3 = 0$$

$$-8i_1 + (13 - j)i_2 - (1 - j3)i_3 = 0 \dots\dots (1)$$

Apply Kirchhoff's voltage law super mesh.

$$-20 \angle 0^\circ + 8(i_1 - i_2) + Z_2 (i_3 - i_2) = 0$$

$$-20 + 8(i_1 - i_2) + (1 - j3)(i_3 - i_2) = 0$$

$$8i_1 - (8 + 1 - j3)i_2 + (1 - j3)i_3 = 20$$

$$8i_1 - (9 - j3)i_2 + (1 - j3)i_3 = 20 \dots\dots (2)$$

Step 3 of 8

From the current source,

$$i_1 - i_3 = 4 \angle -90^\circ$$

$$i_1 - i_3 = -j4 \dots\dots (3)$$

The current, i_3 is the Norton current, I_N .

Solve equations (1), (2) and (3) for Norton current by using Crammer's rule.

$$\begin{bmatrix} -8 & (13-j) & -(1-j3) \\ -1 & 0 & 1 \\ 8 & -(9-j3) & (1-j3) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -j4 \\ 20 \end{bmatrix}$$

Calculate the value of Δ .

$$\begin{aligned} \Delta &= \begin{vmatrix} -8 & 13-j & -1+j3 \\ -1 & 0 & 1 \\ 8 & -9+j3 & 1-j3 \end{vmatrix} \\ &= -8[0 - (-9+j3)] - (13-j)[(-1)(1-j3) - 8] + (-1+j3)[(-1)(-9+j3)] \\ &= -8(9-j3) + (13-j)(9-j3) + (-1+j3)(9-j3) \\ &= (-8+13-j-1+j3)(9-j3) \\ &= (4+j2)(9-j3) \\ &= 42 + j6 \end{aligned}$$

Step 4 of 8

Calculate the value of Δ_3 .

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} -8 & 13-j & 0 \\ -1 & 0 & -j4 \\ 8 & -9+j3 & 20 \end{vmatrix} \\ &= -8[0 - (-j4)(-9+j3)] - (13-j)[(-1)(20) + j32] \\ &= -8(-12 - j36) - (13-j)(-20 + j32) \\ &= 96 + j288 + 228 - j436 \\ \Delta_3 &= 324 - j148\end{aligned}$$

Calculate the Norton current, $\mathbf{I}_N$

$$\begin{aligned}\mathbf{I}_N &= i_3 \\ &= \frac{\Delta_3}{\Delta} \\ &= \frac{324 - j148}{42 + j6} \\ &= 7.0666 - j4.5333\end{aligned}$$

$$\mathbf{I}_N = 8.396 \angle -32.68^\circ$$

Therefore, the Norton current for the circuit, $\mathbf{I}_N$ is $8.396 \angle -32.68^\circ$.

Step 5 of 8

To obtain Norton impedance, $\mathbf{Z}_N$, remove the independent source.

Draw the modified circuit diagram to calculate the Norton impedance, $\mathbf{Z}_N$.

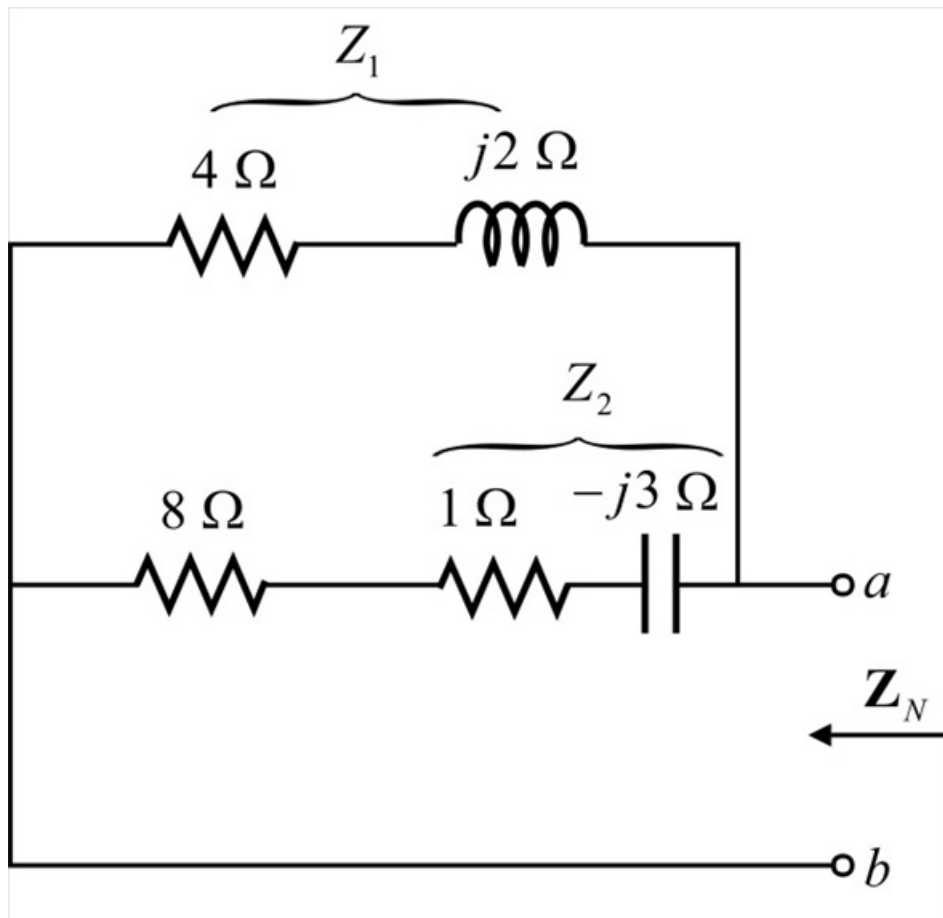


Figure 2

Step 6 of 8

Calculate the Norton's impedance, Z_N .

$$\begin{aligned}
 Z_N &= Z_1 \parallel (8 + Z_2) \\
 &= (4 + j2) \parallel (8 + 1 - j3) \\
 &= (4 + j2) \parallel (9 - j3) \\
 &= \frac{(4 + j2)(9 - j3)}{(4 + j2) + (9 - j3)} \\
 &= \frac{42 + j6}{13 - j} \\
 &= 3.176 + j0.706 \, \Omega
 \end{aligned}$$

Therefore, the Norton impedance for the circuit, Z_N is $3.176 + j0.706 \, \Omega$.

Step 7 of 8

Connect the Norton equivalent to the load and calculate the current, I_o .

Draw the equivalent circuit.

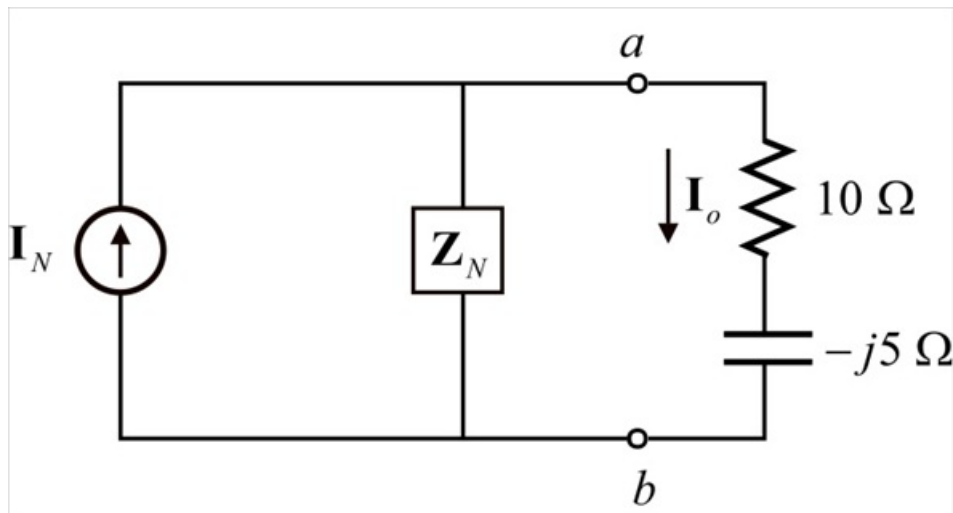


Figure 3

Step 8 of 8

Apply current division rule to calculate the current, I_o .

$$\begin{aligned}
 I_o &= \left(\frac{Z_N}{Z_N + 10 - j5} \right) I_N \\
 &= \left(\frac{3.176 + j0.706}{3.176 + j0.706 + 10 - j5} \right) (8.396 \angle -32.68^\circ) \\
 &= \left(\frac{3.176 + j0.706}{13.176 - j4.294} \right) (8.396 \angle -32.68^\circ) \\
 &= (0.202 + j0.1195) (8.396 \angle -32.68^\circ) \\
 &= (0.2348 \angle 30.58^\circ) (8.396 \angle -32.68^\circ) \\
 &= 1.9714 \angle -2.10^\circ \text{ A}
 \end{aligned}$$

Therefore, the value of the current, I_o is $1.9714 \angle -2.10^\circ \text{ A}$.